In [1]:
```
## ====================================
## EET-4501 – Applied Machine Learning
## Assignment 7
## ====================================
```

In [2]:
```
%pip install -U pip
%pip install tensorflow matplotlib ipykernel pydot
%pip install tensorboard notebook ipykernel
```

```
Requirement already satisfied: pip in /Library/Frameworks/Python.framework/V
ersions/3.11/lib/python3.11/site-packages (26.0.1)
Note: you may need to restart the kernel to use updated packages.
Requirement already satisfied: tensorflow in /Library/Frameworks/Python.fram
ework/Versions/3.11/lib/python3.11/site-packages (2.21.0)
Requirement already satisfied: matplotlib in /Library/Frameworks/Python.fram
ework/Versions/3.11/lib/python3.11/site-packages (3.10.8)
Requirement already satisfied: ipykernel in /Users/athenalieu/Library/Pytho
n/3.11/lib/python/site-packages (7.2.0)
Requirement already satisfied: pydot in /Library/Frameworks/Python.framewor
k/Versions/3.11/lib/python3.11/site-packages (4.0.1)
Requirement already satisfied: absl-py>=1.0.0 in /Library/Frameworks/Python.
framework/Versions/3.11/lib/python3.11/site-packages (from tensorflow) (2.4.
0)
Requirement already satisfied: astunparse>=1.6.0 in /Library/Frameworks/Pyth
on.framework/Versions/3.11/lib/python3.11/site-packages (from tensorflow)
(1.6.3)
Requirement already satisfied: flatbuffers>=25.9.23 in /Library/Frameworks/P
ython.framework/Versions/3.11/lib/python3.11/site-packages (from tensorflow)
(25.12.19)
Requirement already satisfied: gast!=0.5.0,!=0.5.1,!=0.5.2,>=0.2.1 in /Libra
ry/Frameworks/Python.framework/Versions/3.11/lib/python3.11/site-packages (f
rom tensorflow) (0.7.0)
Requirement already satisfied: google_pasta>=0.1.1 in /Library/Frameworks/Py
thon.framework/Versions/3.11/lib/python3.11/site-packages (from tensorflow)
(0.2.0)
Requirement already satisfied: libclang>=13.0.0 in /Library/Frameworks/Pytho
n.framework/Versions/3.11/lib/python3.11/site-packages (from tensorflow) (1
8.1.1)
Requirement already satisfied: opt_einsum>=2.3.2 in /Library/Frameworks/Pyth
on.framework/Versions/3.11/lib/python3.11/site-packages (from tensorflow)
(3.4.0)
Requirement already satisfied: packaging in /Users/athenalieu/Library/Pytho
n/3.11/lib/python/site-packages (from tensorflow) (26.0)
Requirement already satisfied: protobuf<8.0.0,>=6.31.1 in /Library/Framework
s/Python.framework/Versions/3.11/lib/python3.11/site-packages (from tensorfl
ow) (7.34.1)
Requirement already satisfied: requests<3,>=2.21.0 in /Library/Frameworks/Py
thon.framework/Versions/3.11/lib/python3.11/site-packages (from tensorflow)
(2.33.1)
Requirement already satisfied: setuptools in /Library/Frameworks/Python.fram
ework/Versions/3.11/lib/python3.11/site-packages (from tensorflow) (65.5.0)
Requirement already satisfied: six>=1.12.0 in /Users/athenalieu/Library/Pyth
on/3.11/lib/python/site-packages (from tensorflow) (1.17.0)
Requirement already satisfied: termcolor>=1.1.0 in /Library/Frameworks/Pytho
n.framework/Versions/3.11/lib/python3.11/site-packages (from tensorflow) (3.
3.0)
Requirement already satisfied: typing_extensions>=3.6.6 in /Users/athenalie
u/Library/Python/3.11/lib/python/site-packages (from tensorflow) (4.15.0)
Requirement already satisfied: wrapt>=1.11.0 in /Library/Frameworks/Python.f
ramework/Versions/3.11/lib/python3.11/site-packages (from tensorflow) (2.1.
2)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in /Library/Frameworks/Py
thon.framework/Versions/3.11/lib/python3.11/site-packages (from tensorflow)
(1.80.0)
Requirement already satisfied: keras>=3.12.0 in /Library/Frameworks/Python.f
```

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ramework/Versions/3.11/lib/python3.11/site-packages (from tensorflow) (3.14.
0)
Requirement already satisfied: numpy>=1.26.0 in /Library/Frameworks/Python.f
ramework/Versions/3.11/lib/python3.11/site-packages (from tensorflow) (2.4.
4)
Requirement already satisfied: h5py<3.15.0,>=3.11.0 in /Library/Frameworks/P
ython.framework/Versions/3.11/lib/python3.11/site-packages (from tensorflow)
(3.14.0)
Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in /Library/Framework
s/Python.framework/Versions/3.11/lib/python3.11/site-packages (from tensorfl
ow) (0.5.4)
Requirement already satisfied: charset_normalizer<4,>=2 in /Library/Framewor
ks/Python.framework/Versions/3.11/lib/python3.11/site-packages (from request
s<3,>=2.21.0->tensorflow) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in /Library/Frameworks/Python.fr
amework/Versions/3.11/lib/python3.11/site-packages (from requests<3,>=2.21.0
->tensorflow) (3.11)
Requirement already satisfied: urllib3<3,>=1.26 in /Library/Frameworks/Pytho
n.framework/Versions/3.11/lib/python3.11/site-packages (from requests<3,>=2.
21.0->tensorflow) (2.6.3)
Requirement already satisfied: certifi>=2023.5.7 in /Library/Frameworks/Pyth
on.framework/Versions/3.11/lib/python3.11/site-packages (from requests<3,>=
2.21.0->tensorflow) (2026.2.25)
Requirement already satisfied: contourpy>=1.0.1 in /Library/Frameworks/Pytho
n.framework/Versions/3.11/lib/python3.11/site-packages (from matplotlib) (1.
3.3)
Requirement already satisfied: cycler>=0.10 in /Library/Frameworks/Python.fr
amework/Versions/3.11/lib/python3.11/site-packages (from matplotlib) (0.12.
1)
Requirement already satisfied: fonttools>=4.22.0 in /Library/Frameworks/Pyth
on.framework/Versions/3.11/lib/python3.11/site-packages (from matplotlib)
(4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /Library/Frameworks/Pyth
on.framework/Versions/3.11/lib/python3.11/site-packages (from matplotlib)
(1.5.0)
Requirement already satisfied: pillow>=8 in /Library/Frameworks/Python.frame
work/Versions/3.11/lib/python3.11/site-packages (from matplotlib) (12.2.0)
Requirement already satisfied: pyparsing>=3 in /Library/Frameworks/Python.fr
amework/Versions/3.11/lib/python3.11/site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /Users/athenalieu/Lib
rary/Python/3.11/lib/python/site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: appnope>=0.1.2 in /Users/athenalieu/Library/P
ython/3.11/lib/python/site-packages (from ipykernel) (0.1.4)
Requirement already satisfied: comm>=0.1.1 in /Users/athenalieu/Library/Pyth
on/3.11/lib/python/site-packages (from ipykernel) (0.2.3)
Requirement already satisfied: debugpy>=1.6.5 in /Users/athenalieu/Library/P
ython/3.11/lib/python/site-packages (from ipykernel) (1.8.20)
Requirement already satisfied: ipython>=7.23.1 in /Users/athenalieu/Library/
Python/3.11/lib/python/site-packages (from ipykernel) (9.10.1)
Requirement already satisfied: jupyter-client>=8.8.0 in /Users/athenalieu/Li
brary/Python/3.11/lib/python/site-packages (from ipykernel) (8.8.0)
Requirement already satisfied: jupyter-core!=6.0.*,>=5.1 in /Users/athenalie
u/Library/Python/3.11/lib/python/site-packages (from ipykernel) (5.9.1)
Requirement already satisfied: matplotlib-inline>=0.1 in /Users/athenalieu/L
ibrary/Python/3.11/lib/python/site-packages (from ipykernel) (0.2.1)
Requirement already satisfied: nest-asyncio>=1.4 in /Users/athenalieu/Librar
```

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y/Python/3.11/lib/python/site-packages (from ipykernel) (1.6.0)
Requirement already satisfied: psutil>=5.7 in /Users/athenalieu/Library/Pyth
on/3.11/lib/python/site-packages (from ipykernel) (7.2.2)
Requirement already satisfied: pyzmq>=25 in /Users/athenalieu/Library/Pytho
n/3.11/lib/python/site-packages (from ipykernel) (27.1.0)
Requirement already satisfied: tornado>=6.4.1 in /Users/athenalieu/Library/P
ython/3.11/lib/python/site-packages (from ipykernel) (6.5.5)
Requirement already satisfied: traitlets>=5.4.0 in /Users/athenalieu/Librar
y/Python/3.11/lib/python/site-packages (from ipykernel) (5.14.3)
Requirement already satisfied: wheel<1.0,>=0.23.0 in /Library/Frameworks/Pyt
hon.framework/Versions/3.11/lib/python3.11/site-packages (from astunparse>=
1.6.0->tensorflow) (0.46.3)
Requirement already satisfied: decorator>=4.3.2 in /Users/athenalieu/Librar
y/Python/3.11/lib/python/site-packages (from ipython>=7.23.1->ipykernel) (5.
2.1)
Requirement already satisfied: ipython-pygments-lexers>=1.0.0 in /Users/athe
nalieu/Library/Python/3.11/lib/python/site-packages (from ipython>=7.23.1->i
pykernel) (1.1.1)
Requirement already satisfied: jedi>=0.18.1 in /Users/athenalieu/Library/Pyt
hon/3.11/lib/python/site-packages (from ipython>=7.23.1->ipykernel) (0.19.2)
Requirement already satisfied: pexpect>4.3 in /Users/athenalieu/Library/Pyth
on/3.11/lib/python/site-packages (from ipython>=7.23.1->ipykernel) (4.9.0)
Requirement already satisfied: prompt_toolkit<3.1.0,>=3.0.41 in /Users/athen
alieu/Library/Python/3.11/lib/python/site-packages (from ipython>=7.23.1->ip
ykernel) (3.0.52)
Requirement already satisfied: pygments>=2.11.0 in /Users/athenalieu/Librar
y/Python/3.11/lib/python/site-packages (from ipython>=7.23.1->ipykernel) (2.
20.0)
Requirement already satisfied: stack_data>=0.6.0 in /Users/athenalieu/Librar
y/Python/3.11/lib/python/site-packages (from ipython>=7.23.1->ipykernel) (0.
6.3)
Requirement already satisfied: wcwidth in /Users/athenalieu/Library/Python/
3.11/lib/python/site-packages (from prompt_toolkit<3.1.0,>=3.0.41->ipython>=
7.23.1->ipykernel) (0.6.0)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in /Users/athenalieu/Libr
ary/Python/3.11/lib/python/site-packages (from jedi>=0.18.1->ipython>=7.23.1
->ipykernel) (0.8.6)
Requirement already satisfied: platformdirs>=2.5 in /Users/athenalieu/Librar
y/Python/3.11/lib/python/site-packages (from jupyter-core!=6.0.*,>=5.1->ipyk
ernel) (4.9.6)
Requirement already satisfied: rich in /Library/Frameworks/Python.framework/
Versions/3.11/lib/python3.11/site-packages (from keras>=3.12.0->tensorflow)
(15.0.0)
Requirement already satisfied: namex in /Library/Frameworks/Python.framewor
k/Versions/3.11/lib/python3.11/site-packages (from keras>=3.12.0->tensorflo
w) (0.1.0)
Requirement already satisfied: optree in /Library/Frameworks/Python.framewor
k/Versions/3.11/lib/python3.11/site-packages (from keras>=3.12.0->tensorflo
w) (0.19.0)
Requirement already satisfied: ptyprocess>=0.5 in /Users/athenalieu/Library/
Python/3.11/lib/python/site-packages (from pexpect>4.3->ipython>=7.23.1->ipy
kernel) (0.7.0)
Requirement already satisfied: executing>=1.2.0 in /Users/athenalieu/Librar
y/Python/3.11/lib/python/site-packages (from stack_data>=0.6.0->ipython>=7.2
3.1->ipykernel) (2.2.1)
Requirement already satisfied: asttokens>=2.1.0 in /Users/athenalieu/Librar
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y/Python/3.11/lib/python/site-packages (from stack_data>=0.6.0->ipython>=7.2
3.1->ipykernel) (3.0.1)
Requirement already satisfied: pure-eval in /Users/athenalieu/Library/Pytho
n/3.11/lib/python/site-packages (from stack_data>=0.6.0->ipython>=7.23.1->ip
ykernel) (0.2.3)
Requirement already satisfied: markdown-it-py>=2.2.0 in /Library/Frameworks/
Python.framework/Versions/3.11/lib/python3.11/site-packages (from rich->kera
s>=3.12.0->tensorflow) (4.0.0)
Requirement already satisfied: mdurl~=0.1 in /Library/Frameworks/Python.fram
ework/Versions/3.11/lib/python3.11/site-packages (from markdown-it-py>=2.2.0
->rich->keras>=3.12.0->tensorflow) (0.1.2)
Note: you may need to restart the kernel to use updated packages.
Requirement already satisfied: tensorboard in /Library/Frameworks/Python.fra
mework/Versions/3.11/lib/python3.11/site-packages (2.20.0)
Requirement already satisfied: notebook in /Library/Frameworks/Python.framew
ork/Versions/3.11/lib/python3.11/site-packages (7.5.5)
Requirement already satisfied: ipykernel in /Users/athenalieu/Library/Pytho
n/3.11/lib/python/site-packages (7.2.0)
Requirement already satisfied: absl-py>=0.4 in /Library/Frameworks/Python.fr
amework/Versions/3.11/lib/python3.11/site-packages (from tensorboard) (2.4.
0)
Requirement already satisfied: grpcio>=1.48.2 in /Library/Frameworks/Python.
framework/Versions/3.11/lib/python3.11/site-packages (from tensorboard) (1.8
0.0)
Requirement already satisfied: markdown>=2.6.8 in /Library/Frameworks/Pytho
n.framework/Versions/3.11/lib/python3.11/site-packages (from tensorboard)
(3.10.2)
Requirement already satisfied: numpy>=1.12.0 in /Library/Frameworks/Python.f
ramework/Versions/3.11/lib/python3.11/site-packages (from tensorboard) (2.4.
4)
Requirement already satisfied: packaging in /Users/athenalieu/Library/Pytho
n/3.11/lib/python/site-packages (from tensorboard) (26.0)
Requirement already satisfied: pillow in /Library/Frameworks/Python.framewor
k/Versions/3.11/lib/python3.11/site-packages (from tensorboard) (12.2.0)
Requirement already satisfied: protobuf!=4.24.0,>=3.19.6 in /Library/Framewo
rks/Python.framework/Versions/3.11/lib/python3.11/site-packages (from tensor
board) (7.34.1)
Requirement already satisfied: setuptools>=41.0.0 in /Library/Frameworks/Pyt
hon.framework/Versions/3.11/lib/python3.11/site-packages (from tensorboard)
(65.5.0)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in /Lib
rary/Frameworks/Python.framework/Versions/3.11/lib/python3.11/site-packages
(from tensorboard) (0.7.2)
Requirement already satisfied: werkzeug>=1.0.1 in /Library/Frameworks/Pytho
n.framework/Versions/3.11/lib/python3.11/site-packages (from tensorboard)
(3.1.8)
Requirement already satisfied: jupyter-server<3,>=2.4.0 in /Library/Framewor
ks/Python.framework/Versions/3.11/lib/python3.11/site-packages (from noteboo
k) (2.17.0)
Requirement already satisfied: jupyterlab-server<3,>=2.28.0 in /Library/Fram
eworks/Python.framework/Versions/3.11/lib/python3.11/site-packages (from not
ebook) (2.28.0)
Requirement already satisfied: jupyterlab<4.6,>=4.5.6 in /Library/Framework
s/Python.framework/Versions/3.11/lib/python3.11/site-packages (from noteboo
k) (4.5.6)
Requirement already satisfied: notebook-shim<0.3,>=0.2 in /Library/Framework
```

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s/Python.framework/Versions/3.11/lib/python3.11/site-packages (from noteboo
k) (0.2.4)
Requirement already satisfied: tornado>=6.2.0 in /Users/athenalieu/Library/P
ython/3.11/lib/python/site-packages (from notebook) (6.5.5)
Requirement already satisfied: anyio>=3.1.0 in /Library/Frameworks/Python.fr
amework/Versions/3.11/lib/python3.11/site-packages (from jupyter-server<3,>=
2.4.0->notebook) (4.13.0)
Requirement already satisfied: argon2-cffi>=21.1 in /Library/Frameworks/Pyth
on.framework/Versions/3.11/lib/python3.11/site-packages (from jupyter-server
<3,>=2.4.0->notebook) (25.1.0)
Requirement already satisfied: jinja2>=3.0.3 in /Library/Frameworks/Python.f
ramework/Versions/3.11/lib/python3.11/site-packages (from jupyter-server<3,>
=2.4.0->notebook) (3.1.6)
Requirement already satisfied: jupyter-client>=7.4.4 in /Users/athenalieu/Li
brary/Python/3.11/lib/python/site-packages (from jupyter-server<3,>=2.4.0->n
otebook) (8.8.0)
Requirement already satisfied: jupyter-core!=5.0.*,>=4.12 in /Users/athenali
eu/Library/Python/3.11/lib/python/site-packages (from jupyter-server<3,>=2.
4.0->notebook) (5.9.1)
Requirement already satisfied: jupyter-events>=0.11.0 in /Library/Framework
s/Python.framework/Versions/3.11/lib/python3.11/site-packages (from jupyter-
server<3,>=2.4.0->notebook) (0.12.0)
Requirement already satisfied: jupyter-server-terminals>=0.4.4 in /Library/F
rameworks/Python.framework/Versions/3.11/lib/python3.11/site-packages (from
jupyter-server<3,>=2.4.0->notebook) (0.5.4)
Requirement already satisfied: nbconvert>=6.4.4 in /Library/Frameworks/Pytho
n.framework/Versions/3.11/lib/python3.11/site-packages (from jupyter-server<
3,>=2.4.0->notebook) (7.17.1)
Requirement already satisfied: nbformat>=5.3.0 in /Library/Frameworks/Pytho
n.framework/Versions/3.11/lib/python3.11/site-packages (from jupyter-server<
3,>=2.4.0->notebook) (5.10.4)
Requirement already satisfied: overrides>=5.0 in /Library/Frameworks/Python.
framework/Versions/3.11/lib/python3.11/site-packages (from jupyter-server<3,
>=2.4.0->notebook) (7.7.0)
Requirement already satisfied: prometheus-client>=0.9 in /Library/Framework
s/Python.framework/Versions/3.11/lib/python3.11/site-packages (from jupyter-
server<3,>=2.4.0->notebook) (0.25.0)
Requirement already satisfied: pyzmq>=24 in /Users/athenalieu/Library/Pytho
n/3.11/lib/python/site-packages (from jupyter-server<3,>=2.4.0->notebook) (2
7.1.0)
Requirement already satisfied: send2trash>=1.8.2 in /Library/Frameworks/Pyth
on.framework/Versions/3.11/lib/python3.11/site-packages (from jupyter-server
<3,>=2.4.0->notebook) (2.1.0)
Requirement already satisfied: terminado>=0.8.3 in /Library/Frameworks/Pytho
n.framework/Versions/3.11/lib/python3.11/site-packages (from jupyter-server<
3,>=2.4.0->notebook) (0.18.1)
Requirement already satisfied: traitlets>=5.6.0 in /Users/athenalieu/Librar
y/Python/3.11/lib/python/site-packages (from jupyter-server<3,>=2.4.0->noteb
ook) (5.14.3)
Requirement already satisfied: websocket-client>=1.7 in /Library/Frameworks/
Python.framework/Versions/3.11/lib/python3.11/site-packages (from jupyter-se
rver<3,>=2.4.0->notebook) (1.9.0)
Requirement already satisfied: async-lru>=1.0.0 in /Library/Frameworks/Pytho
n.framework/Versions/3.11/lib/python3.11/site-packages (from jupyterlab<4.6,
>=4.5.6->notebook) (2.3.0)
Requirement already satisfied: httpx<1,>=0.25.0 in /Library/Frameworks/Pytho
```

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n.framework/Versions/3.11/lib/python3.11/site-packages (from jupyterlab<4.6,
>=4.5.6->notebook) (0.28.1)
Requirement already satisfied: jupyter-lsp>=2.0.0 in /Library/Frameworks/Pyt
hon.framework/Versions/3.11/lib/python3.11/site-packages (from jupyterlab<4.
6,>=4.5.6->notebook) (2.3.1)
Requirement already satisfied: certifi in /Library/Frameworks/Python.framewo
rk/Versions/3.11/lib/python3.11/site-packages (from httpx<1,>=0.25.0->jupyte
rlab<4.6,>=4.5.6->notebook) (2026.2.25)
Requirement already satisfied: httpcore==1.* in /Library/Frameworks/Python.f
ramework/Versions/3.11/lib/python3.11/site-packages (from httpx<1,>=0.25.0->
jupyterlab<4.6,>=4.5.6->notebook) (1.0.9)
Requirement already satisfied: idna in /Library/Frameworks/Python.framework/
Versions/3.11/lib/python3.11/site-packages (from httpx<1,>=0.25.0->jupyterla
b<4.6,>=4.5.6->notebook) (3.11)
Requirement already satisfied: h11>=0.16 in /Library/Frameworks/Python.frame
work/Versions/3.11/lib/python3.11/site-packages (from httpcore==1.*->httpx<
1,>=0.25.0->jupyterlab<4.6,>=4.5.6->notebook) (0.16.0)
Requirement already satisfied: babel>=2.10 in /Library/Frameworks/Python.fra
mework/Versions/3.11/lib/python3.11/site-packages (from jupyterlab-server<3,
>=2.28.0->notebook) (2.18.0)
Requirement already satisfied: json5>=0.9.0 in /Library/Frameworks/Python.fr
amework/Versions/3.11/lib/python3.11/site-packages (from jupyterlab-server<
3,>=2.28.0->notebook) (0.14.0)
Requirement already satisfied: jsonschema>=4.18.0 in /Library/Frameworks/Pyt
hon.framework/Versions/3.11/lib/python3.11/site-packages (from jupyterlab-se
rver<3,>=2.28.0->notebook) (4.26.0)
Requirement already satisfied: requests>=2.31 in /Library/Frameworks/Python.
framework/Versions/3.11/lib/python3.11/site-packages (from jupyterlab-server
<3,>=2.28.0->notebook) (2.33.1)
Requirement already satisfied: appnope>=0.1.2 in /Users/athenalieu/Library/P
ython/3.11/lib/python/site-packages (from ipykernel) (0.1.4)
Requirement already satisfied: comm>=0.1.1 in /Users/athenalieu/Library/Pyth
on/3.11/lib/python/site-packages (from ipykernel) (0.2.3)
Requirement already satisfied: debugpy>=1.6.5 in /Users/athenalieu/Library/P
ython/3.11/lib/python/site-packages (from ipykernel) (1.8.20)
Requirement already satisfied: ipython>=7.23.1 in /Users/athenalieu/Library/
Python/3.11/lib/python/site-packages (from ipykernel) (9.10.1)
Requirement already satisfied: matplotlib-inline>=0.1 in /Users/athenalieu/L
ibrary/Python/3.11/lib/python/site-packages (from ipykernel) (0.2.1)
Requirement already satisfied: nest-asyncio>=1.4 in /Users/athenalieu/Librar
y/Python/3.11/lib/python/site-packages (from ipykernel) (1.6.0)
Requirement already satisfied: psutil>=5.7 in /Users/athenalieu/Library/Pyth
on/3.11/lib/python/site-packages (from ipykernel) (7.2.2)
Requirement already satisfied: typing_extensions>=4.5 in /Users/athenalieu/L
ibrary/Python/3.11/lib/python/site-packages (from anyio>=3.1.0->jupyter-serv
er<3,>=2.4.0->notebook) (4.15.0)
Requirement already satisfied: argon2-cffi-bindings in /Library/Frameworks/P
ython.framework/Versions/3.11/lib/python3.11/site-packages (from argon2-cffi
>=21.1->jupyter-server<3,>=2.4.0->notebook) (25.1.0)
Requirement already satisfied: decorator>=4.3.2 in /Users/athenalieu/Librar
y/Python/3.11/lib/python/site-packages (from ipython>=7.23.1->ipykernel) (5.
2.1)
Requirement already satisfied: ipython-pygments-lexers>=1.0.0 in /Users/athe
nalieu/Library/Python/3.11/lib/python/site-packages (from ipython>=7.23.1->i
pykernel) (1.1.1)
Requirement already satisfied: jedi>=0.18.1 in /Users/athenalieu/Library/Pyt
```

```
hon/3.11/lib/python/site-packages (from ipython>=7.23.1->ipykernel) (0.19.2)
Requirement already satisfied: pexpect>4.3 in /Users/athenalieu/Library/Pyth
on/3.11/lib/python/site-packages (from ipython>=7.23.1->ipykernel) (4.9.0)
Requirement already satisfied: prompt_toolkit<3.1.0,>=3.0.41 in /Users/athen
alieu/Library/Python/3.11/lib/python/site-packages (from ipython>=7.23.1->ip
ykernel) (3.0.52)
Requirement already satisfied: pygments>=2.11.0 in /Users/athenalieu/Librar
y/Python/3.11/lib/python/site-packages (from ipython>=7.23.1->ipykernel) (2.
20.0)
Requirement already satisfied: stack_data>=0.6.0 in /Users/athenalieu/Librar
y/Python/3.11/lib/python/site-packages (from ipython>=7.23.1->ipykernel) (0.
6.3)
Requirement already satisfied: wcwidth in /Users/athenalieu/Library/Python/
3.11/lib/python/site-packages (from prompt_toolkit<3.1.0,>=3.0.41->ipython>=
7.23.1->ipykernel) (0.6.0)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in /Users/athenalieu/Libr
ary/Python/3.11/lib/python/site-packages (from jedi>=0.18.1->ipython>=7.23.1
->ipykernel) (0.8.6)
Requirement already satisfied: MarkupSafe>=2.0 in /Library/Frameworks/Pytho
n.framework/Versions/3.11/lib/python3.11/site-packages (from jinja2>=3.0.3->
jupyter-server<3,>=2.4.0->notebook) (3.0.3)
Requirement already satisfied: attrs>=22.2.0 in /Library/Frameworks/Python.f
ramework/Versions/3.11/lib/python3.11/site-packages (from jsonschema>=4.18.0
->jupyterlab-server<3,>=2.28.0->notebook) (26.1.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /Libr
ary/Frameworks/Python.framework/Versions/3.11/lib/python3.11/site-packages
(from jsonschema>=4.18.0->jupyterlab-server<3,>=2.28.0->notebook) (2025.9.1)
Requirement already satisfied: referencing>=0.28.4 in /Library/Frameworks/Py
thon.framework/Versions/3.11/lib/python3.11/site-packages (from jsonschema>=
4.18.0->jupyterlab-server<3,>=2.28.0->notebook) (0.37.0)
Requirement already satisfied: rpds-py>=0.25.0 in /Library/Frameworks/Pytho
n.framework/Versions/3.11/lib/python3.11/site-packages (from jsonschema>=4.1
8.0->jupyterlab-server<3,>=2.28.0->notebook) (0.30.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /Users/athenalieu/L
ibrary/Python/3.11/lib/python/site-packages (from jupyter-client>=7.4.4->jup
yter-server<3,>=2.4.0->notebook) (2.9.0.post0)
Requirement already satisfied: platformdirs>=2.5 in /Users/athenalieu/Librar
y/Python/3.11/lib/python/site-packages (from jupyter-core!=5.0.*,>=4.12->jup
yter-server<3,>=2.4.0->notebook) (4.9.6)
Requirement already satisfied: python-json-logger>=2.0.4 in /Library/Framewo
rks/Python.framework/Versions/3.11/lib/python3.11/site-packages (from jupyte
r-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook) (4.1.0)
Requirement already satisfied: pyyaml>=5.3 in /Library/Frameworks/Python.fra
mework/Versions/3.11/lib/python3.11/site-packages (from jupyter-events>=0.1
1.0->jupyter-server<3,>=2.4.0->notebook) (6.0.3)
Requirement already satisfied: rfc3339-validator in /Library/Frameworks/Pyth
on.framework/Versions/3.11/lib/python3.11/site-packages (from jupyter-events
>=0.11.0->jupyter-server<3,>=2.4.0->notebook) (0.1.4)
Requirement already satisfied: rfc3986-validator>=0.1.1 in /Library/Framewor
ks/Python.framework/Versions/3.11/lib/python3.11/site-packages (from jupyter
-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook) (0.1.1)
Requirement already satisfied: fqdn in /Library/Frameworks/Python.framework/
Versions/3.11/lib/python3.11/site-packages (from jsonschema[format-nongpl]>=
4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook) (1.5.1)
Requirement already satisfied: isoduration in /Library/Frameworks/Python.fra
mework/Versions/3.11/lib/python3.11/site-packages (from jsonschema[format-no
```

```
ngpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook)
(20.11.0)
Requirement already satisfied: jsonpointer>1.13 in /Library/Frameworks/Pytho
n.framework/Versions/3.11/lib/python3.11/site-packages (from jsonschema[form
at-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebo
ok) (3.1.1)
Requirement already satisfied: rfc3987-syntax>=1.1.0 in /Library/Frameworks/
Python.framework/Versions/3.11/lib/python3.11/site-packages (from jsonschema
[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->n
otebook) (1.1.0)
Requirement already satisfied: uri-template in /Library/Frameworks/Python.fr
amework/Versions/3.11/lib/python3.11/site-packages (from jsonschema[format-n
ongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->notebook)
(1.3.0)
Requirement already satisfied: webcolors>=24.6.0 in /Library/Frameworks/Pyth
on.framework/Versions/3.11/lib/python3.11/site-packages (from jsonschema[for
mat-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->noteb
ook) (25.10.0)
Requirement already satisfied: beautifulsoup4 in /Library/Frameworks/Python.
framework/Versions/3.11/lib/python3.11/site-packages (from nbconvert>=6.4.4-
>jupyter-server<3,>=2.4.0->notebook) (4.14.3)
Requirement already satisfied: bleach!=5.0.0 in /Library/Frameworks/Python.f
ramework/Versions/3.11/lib/python3.11/site-packages (from bleach[css]!=5.0.0
->nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook) (6.3.0)
Requirement already satisfied: defusedxml in /Library/Frameworks/Python.fram
ework/Versions/3.11/lib/python3.11/site-packages (from nbconvert>=6.4.4->jup
yter-server<3,>=2.4.0->notebook) (0.7.1)
Requirement already satisfied: jupyterlab-pygments in /Library/Frameworks/Py
thon.framework/Versions/3.11/lib/python3.11/site-packages (from nbconvert>=
6.4.4->jupyter-server<3,>=2.4.0->notebook) (0.3.0)
Requirement already satisfied: mistune<4,>=2.0.3 in /Library/Frameworks/Pyth
on.framework/Versions/3.11/lib/python3.11/site-packages (from nbconvert>=6.
4.4->jupyter-server<3,>=2.4.0->notebook) (3.2.0)
Requirement already satisfied: nbclient>=0.5.0 in /Library/Frameworks/Pytho
n.framework/Versions/3.11/lib/python3.11/site-packages (from nbconvert>=6.4.
4->jupyter-server<3,>=2.4.0->notebook) (0.10.4)
Requirement already satisfied: pandocfilters>=1.4.1 in /Library/Frameworks/P
ython.framework/Versions/3.11/lib/python3.11/site-packages (from nbconvert>=
6.4.4->jupyter-server<3,>=2.4.0->notebook) (1.5.1)
Requirement already satisfied: webencodings in /Library/Frameworks/Python.fr
amework/Versions/3.11/lib/python3.11/site-packages (from bleach!=5.0.0->blea
ch[css]!=5.0.0->nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook) (0.5.
1)
Requirement already satisfied: tinycss2<1.5,>=1.1.0 in /Library/Frameworks/P
ython.framework/Versions/3.11/lib/python3.11/site-packages (from bleach[cs
s]!=5.0.0->nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook) (1.4.0)
Requirement already satisfied: fastjsonschema>=2.15 in /Library/Frameworks/P
ython.framework/Versions/3.11/lib/python3.11/site-packages (from nbformat>=
5.3.0->jupyter-server<3,>=2.4.0->notebook) (2.21.2)
Requirement already satisfied: ptyprocess>=0.5 in /Users/athenalieu/Library/
Python/3.11/lib/python/site-packages (from pexpect>4.3->ipython>=7.23.1->ipy
kernel) (0.7.0)
Requirement already satisfied: six>=1.5 in /Users/athenalieu/Library/Python/
3.11/lib/python/site-packages (from python-dateutil>=2.8.2->jupyter-client>=
7.4.4->jupyter-server<3,>=2.4.0->notebook) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /Library/Framewor
```

```
ks/Python.framework/Versions/3.11/lib/python3.11/site-packages (from request
s>=2.31->jupyterlab-server<3,>=2.28.0->notebook) (3.4.7)
Requirement already satisfied: urllib3<3,>=1.26 in /Library/Frameworks/Pytho
n.framework/Versions/3.11/lib/python3.11/site-packages (from requests>=2.31-
>jupyterlab-server<3,>=2.28.0->notebook) (2.6.3)
Requirement already satisfied: lark>=1.2.2 in /Library/Frameworks/Python.fra
mework/Versions/3.11/lib/python3.11/site-packages (from rfc3987-syntax>=1.1.
0->jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server
<3,>=2.4.0->notebook) (1.3.1)
Requirement already satisfied: executing>=1.2.0 in /Users/athenalieu/Librar
y/Python/3.11/lib/python/site-packages (from stack_data>=0.6.0->ipython>=7.2
3.1->ipykernel) (2.2.1)
Requirement already satisfied: asttokens>=2.1.0 in /Users/athenalieu/Librar
y/Python/3.11/lib/python/site-packages (from stack_data>=0.6.0->ipython>=7.2
3.1->ipykernel) (3.0.1)
Requirement already satisfied: pure-eval in /Users/athenalieu/Library/Pytho
n/3.11/lib/python/site-packages (from stack_data>=0.6.0->ipython>=7.23.1->ip
ykernel) (0.2.3)
Requirement already satisfied: cffi>=1.0.1 in /Library/Frameworks/Python.fra
mework/Versions/3.11/lib/python3.11/site-packages (from argon2-cffi-bindings
->argon2-cffi>=21.1->jupyter-server<3,>=2.4.0->notebook) (2.0.0)
Requirement already satisfied: pycparser in /Library/Frameworks/Python.frame
work/Versions/3.11/lib/python3.11/site-packages (from cffi>=1.0.1->argon2-cf
fi-bindings->argon2-cffi>=21.1->jupyter-server<3,>=2.4.0->notebook) (3.0)
Requirement already satisfied: soupsieve>=1.6.1 in /Library/Frameworks/Pytho
n.framework/Versions/3.11/lib/python3.11/site-packages (from beautifulsoup4-
>nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook) (2.8.3)
Requirement already satisfied: arrow>=0.15.0 in /Library/Frameworks/Python.f
ramework/Versions/3.11/lib/python3.11/site-packages (from isoduration->jsons
chema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.
4.0->notebook) (1.4.0)
Requirement already satisfied: tzdata in /Library/Frameworks/Python.framewor
k/Versions/3.11/lib/python3.11/site-packages (from arrow>=0.15.0->isoduratio
n->jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server
<3,>=2.4.0->notebook) (2026.1)
Note: you may need to restart the kernel to use updated packages.
```

In [3]:
```
import tensorflow as tf
import sys
import numpy as np
import matplotlib.pyplot as plt
import datetime

from tensorflow.keras.datasets import mnist
from tensorflow.keras.utils import plot_model
```

In [4]:
```
print("TensorFlow version:", tf.__version__)
print("Python path:", sys.executable)
```

```
TensorFlow version: 2.21.0
Python path: /usr/local/bin/python3
```

In [5]:
```
try:
    import tensorboard
    tensorboard_available = True
    print("TensorBoard is installed.")
```

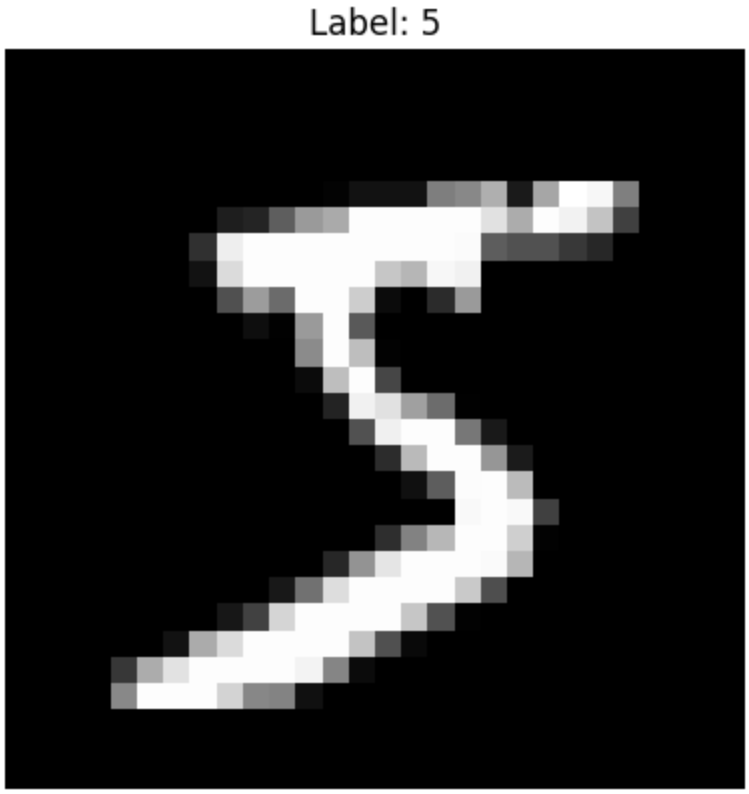
```
except ImportError:
    tensorboard_available = False
    print("TensorBoard is NOT installed. Training will continue without Tens
```

```
TensorBoard is installed.
```

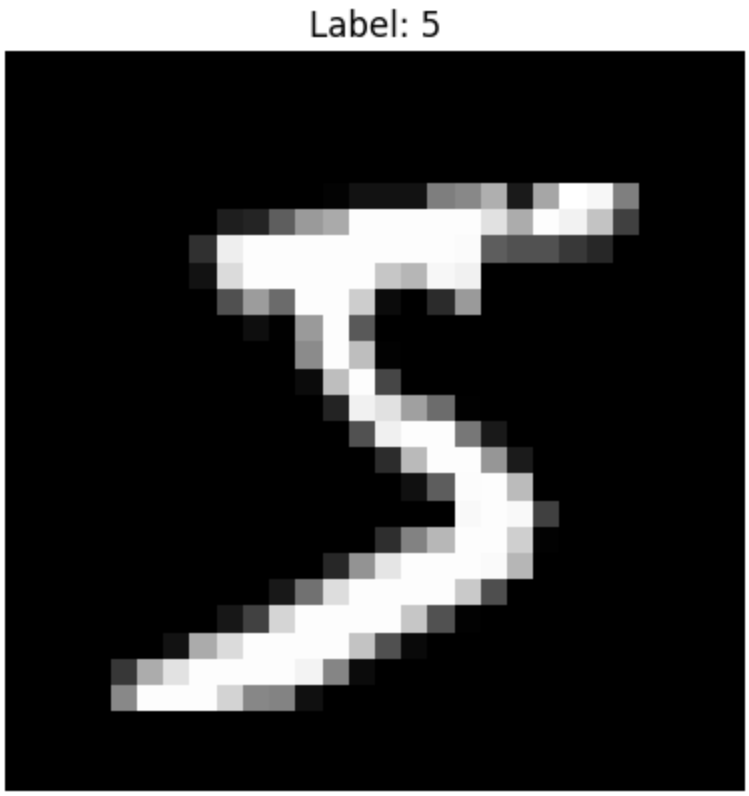
In [6]:

```
## Part 1: Exploring the Data

from tensorflow.keras.datasets import mnist
(x_train, y_train), (x_test, y_test) = mnist.load_data()
print(x_train.shape, x_test.shape)

print("Training images:", x_train.shape[0])
print("Testing images:", x_test.shape[0])

num_classes = len(np.unique(y_train))
print("Number of classes:", num_classes)

def show_image(image, label):
    plt.imshow(image, cmap='gray')
    plt.title(f"Label: {label}")
    plt.axis('off')
    plt.show()

## Display one image from the training set
show_image(x_train[0], y_train[0])

print("Shape of displayed image:", x_train[0].shape)

## Use astype("float32") for safer TensorFlow training
x_train_norm = x_train.astype("float32") / 255.0
x_test_norm = x_test.astype("float32") / 255.0

## Display the same image after normalization
show_image(x_train_norm[0], y_train[0])

## The MNIST dataset contains 60,000 training images and 10,000 testing imag
## The dataset contains 10 classes, representing digits 0 through 9.
## The shape of each image is (28, 28).
## The dataset was normalized by dividing pixel values by 255 so that all va
```

```
(60000, 28, 28) (10000, 28, 28)
Training images: 60000
Testing images: 10000
Number of classes: 10
```




```
Shape of displayed image: (28, 28)
```



In [7]:
```
## Part 2: Build the Neural Network Model

## Build the model through a function so it is easier to create two separate
def build_model():
    model = tf.keras.Sequential([
```

```
        tf.keras.layers.Flatten(input_shape=(28, 28)),
        tf.keras.layers.Dense(128, activation='tanh'),
        tf.keras.layers.Dense(128, activation='tanh'),
        tf.keras.layers.Dense(128, activation='tanh'),
        tf.keras.layers.Dense(num_classes)  # Output layer
    ])
    return model

model = build_model()
model.summary()

try:
    plot_model(model, to_file='Assignment7_model.png', show_shapes=True, sho
    print
except Exception as e:
    print(f"Error occurred while plotting the model: {e}")
```

```
/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/site-packa
ges/keras/src/layers/reshaping/flatten.py:37: UserWarning: Do not pass an `i
nput_shape`/`input_dim` argument to a layer. When using Sequential models, p
refer using an `Input(shape)` object as the first layer in the model instea
d.
  super().__init__(**kwargs)
```

Model: "sequential"

Layer (type)	Output Shape	Par
flatten (Flatten)	(None, 784)	
dense (Dense)	(None, 128)	100
dense_1 (Dense)	(None, 128)	16
dense_2 (Dense)	(None, 128)	16
dense_3 (Dense)	(None, 10)	1

Total params: 134,794 (526.54 KB)

Trainable params: 134,794 (526.54 KB)

Non-trainable params: 0 (0.00 B)

```
You must install graphviz (see instructions at https://graphviz.gitlab.io/do
wnload/) for `plot_model` to work.
```

In [8]:

```
## Part 3: Model Configuration

model_adam = tf.keras.models.clone_model(model)

model_adam.compile(
    optimizer='adam',
    loss=tf.keras.losses.SparseCategoricalCrossentropy(from_logits=True),
    metrics=['accuracy']
)

model_rmsprop = build_model()
```

```
model_rmsprop.compile(
    optimizer='rmsprop',
    loss=tf.keras.losses.SparseCategoricalCrossentropy(from_logits=True),
    metrics=['accuracy']
)
```

```
/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/site-packa
ges/keras/src/layers/reshaping/flatten.py:37: UserWarning: Do not pass an `i
nput_shape`/`input_dim` argument to a layer. When using Sequential models, p
refer using an `Input(shape)` object as the first layer in the model instea
d.
  super().__init__(**kwargs)
```

In [9]:

```
## Part 4: Model Training

log_dir_adam = "logs/fit/adam_" + datetime.datetime.now().strftime("%Y%m%d-%
print("Adam log directory:", log_dir_adam)

callbacks_adam = []

if tensorboard_available:
    callbacks_adam.append(
        tf.keras.callbacks.TensorBoard(
            log_dir=log_dir_adam,
            histogram_freq=0
    )
)

history_adam = model_adam.fit(
    x_train_norm, y_train,
    epochs=10,
    batch_size=32,
    validation_data=(x_test_norm, y_test),
    callbacks=callbacks_adam,
    verbose=1
)

log_dir_rmsprop = "logs/fit/rmsprop_" + datetime.datetime.now().strftime("%Y
print("RMSprop log directory:", log_dir_rmsprop)

callbacks_rmsprop = []

if tensorboard_available:
    callbacks_rmsprop.append(
        tf.keras.callbacks.TensorBoard(
            log_dir=log_dir_rmsprop,
            histogram_freq=0
    )
)

history_rmsprop = model_rmsprop.fit(
    x_train_norm, y_train,
    epochs=10,
    batch_size=32,
    validation_data=(x_test_norm, y_test),
    callbacks=callbacks_rmsprop,
```

```
    verbose=1
)

%load_ext tensorboard
%tensorboard --logdir logs/fit

adam_final_acc = history_adam.history['val_accuracy'][-1]
rmsprop_final_acc = history_rmsprop.history['val_accuracy'][-1]

print("Final validation accuracy (Adam):", adam_final_acc)
print("Final validation accuracy (RMSprop):", rmsprop_final_acc)

if adam_final_acc > rmsprop_final_acc:
    print("Adam performed better than RMSprop.")
else:
    print("RMSprop performed better than Adam.")

print("Explanation:")
print("Adam often performs better because it adapts the learning rate for ea
print("which can lead to faster and more stable convergence during training.

## The optimizer with better accuracy was the one that had the higher final
## In many cases, Adam performs better as it adapts the learning rate during
```

```
Adam log directory: logs/fit/adam_20260413-183645
Epoch 1/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 3s 1ms/step - accuracy: 0.9256 - loss: 0.2463
- val_accuracy: 0.9563 - val_loss: 0.1379
Epoch 2/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 3s 1ms/step - accuracy: 0.9631 - loss: 0.1175
- val_accuracy: 0.9700 - val_loss: 0.0979
Epoch 3/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 3s 1ms/step - accuracy: 0.9743 - loss: 0.0829
- val_accuracy: 0.9718 - val_loss: 0.0923
Epoch 4/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 3s 1ms/step - accuracy: 0.9819 - loss: 0.0589
- val_accuracy: 0.9742 - val_loss: 0.0902
Epoch 5/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 3s 1ms/step - accuracy: 0.9844 - loss: 0.0477
- val_accuracy: 0.9734 - val_loss: 0.0915
Epoch 6/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 3s 1ms/step - accuracy: 0.9879 - loss: 0.0370
- val_accuracy: 0.9749 - val_loss: 0.0866
Epoch 7/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 3s 1ms/step - accuracy: 0.9896 - loss: 0.0320
- val_accuracy: 0.9754 - val_loss: 0.0936
Epoch 8/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 3s 1ms/step - accuracy: 0.9915 - loss: 0.0258
- val_accuracy: 0.9738 - val_loss: 0.0938
Epoch 9/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 3s 1ms/step - accuracy: 0.9923 - loss: 0.0241
- val_accuracy: 0.9785 - val_loss: 0.0802
Epoch 10/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 3s 1ms/step - accuracy: 0.9934 - loss: 0.0196
- val_accuracy: 0.9770 - val_loss: 0.0927
RMSprop log directory: logs/fit/rmsprop_20260413-183713
Epoch 1/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 3s 1ms/step - accuracy: 0.9240 - loss: 0.2506
- val_accuracy: 0.9545 - val_loss: 0.1496
Epoch 2/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 2s 1ms/step - accuracy: 0.9632 - loss: 0.1176
- val_accuracy: 0.9670 - val_loss: 0.1111
Epoch 3/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 2s 1ms/step - accuracy: 0.9751 - loss: 0.0820
- val_accuracy: 0.9693 - val_loss: 0.1042
Epoch 4/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 2s 1ms/step - accuracy: 0.9801 - loss: 0.0628
- val_accuracy: 0.9703 - val_loss: 0.0956
Epoch 5/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 2s 1ms/step - accuracy: 0.9850 - loss: 0.0487
- val_accuracy: 0.9756 - val_loss: 0.0807
Epoch 6/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 2s 1ms/step - accuracy: 0.9877 - loss: 0.0381
- val_accuracy: 0.9745 - val_loss: 0.0904
Epoch 7/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 3s 2ms/step - accuracy: 0.9904 - loss: 0.0305
- val_accuracy: 0.9740 - val_loss: 0.1019
Epoch 8/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 2s 795us/step - accuracy: 0.9924 - loss: 0.02
43 - val_accuracy: 0.9745 - val_loss: 0.1005
```

```
Epoch 9/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 1s 694us/step - accuracy: 0.9942 - loss: 0.01
91 - val_accuracy: 0.9778 - val_loss: 0.0907
Epoch 10/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 1s 702us/step - accuracy: 0.9952 - loss: 0.01
51 - val_accuracy: 0.9785 - val_loss: 0.0982
Reusing TensorBoard on port 6006 (pid 79158), started 0:12:13 ago. (Use '!ki
ll 79158' to kill it.)
```

```
Final validation accuracy (Adam): 0.9769999980926514
Final validation accuracy (RMSprop): 0.9785000085830688
RMSprop performed better than Adam.
Explanation:
Adam often performs better because it adapts the learning rate for each para
meter,
which can lead to faster and more stable convergence during training.
```

In [10]:

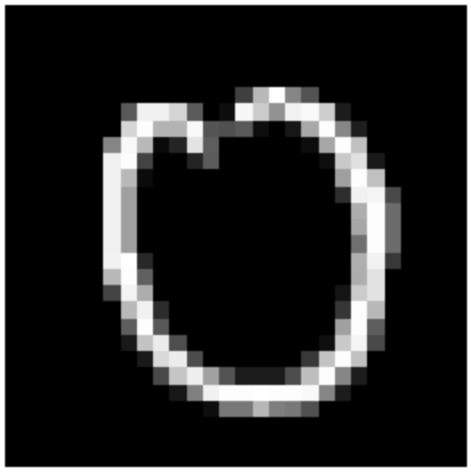
```
## Part 5: Evaluation and Analysis

test_loss_adam, test_acc_adam = model_adam.evaluate(x_test_norm, y_test, ver
test_loss_rms, test_acc_rms = model_rmsprop.evaluate(x_test_norm, y_test, ve

print("Adam Test Loss:", test_loss_adam)
print("Adam Test Accuracy:", test_acc_adam)

print("RMSprop Test Loss:", test_loss_rms)
print("RMSprop Test Accuracy:", test_acc_rms)
```

```
Adam Test Loss: 0.09273218363523483
Adam Test Accuracy: 0.9769999980926514
RMSprop Test Loss: 0.09824065864086151
RMSprop Test Accuracy: 0.9785000085830688
```

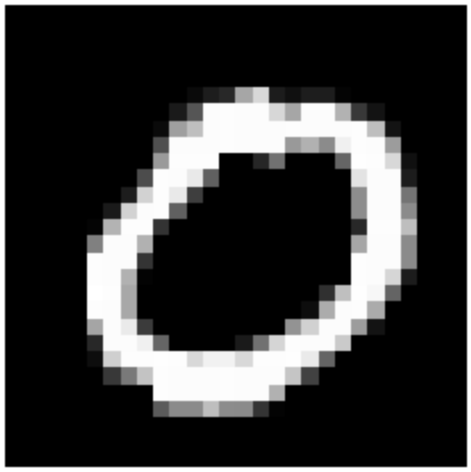
In [11]:

```
## Part 6: Model Testing

## Softmax for probabilities
prob_model = tf.keras.Sequential([
    model_adam,
    tf.keras.layers.Softmax()
])

## Select 10 test images
indices = np.random.choice(len(x_test), 10, replace=False)

for i in indices:
    img = x_test_norm[i]
    true_label = y_test[i]

    prediction = prob_model.predict(img.reshape(1,28,28), verbose=0)
    predicted_label = np.argmax(prediction)
    confidence = np.max(prediction)

    plt.figure(figsize=(3,3))
    plt.imshow(img, cmap='gray')
    plt.title(f"Pred: {predicted_label} | True: {true_label} | Confidence: {
    plt.axis('off')
    plt.show()
```

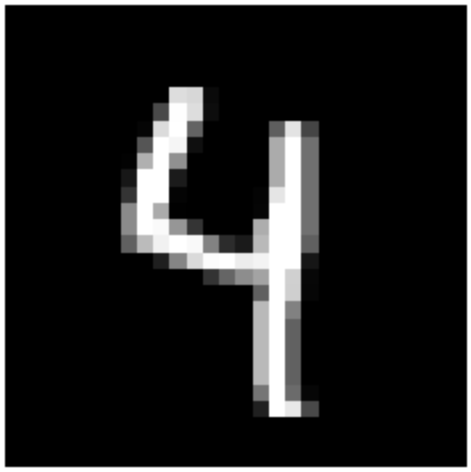
Pred: 0 | True: 0 | Confidence: 1.0000



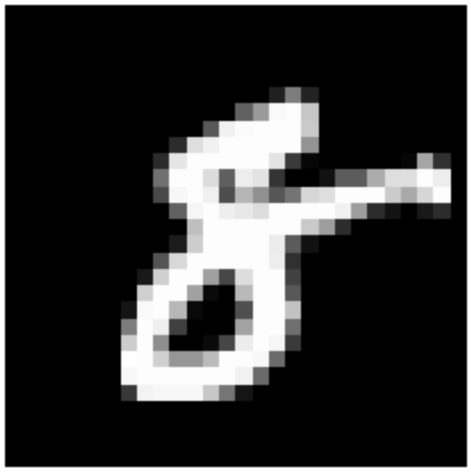
Pred: 0 | True: 0 | Confidence: 1.0000



Pred: 4 | True: 4 | Confidence: 0.9997



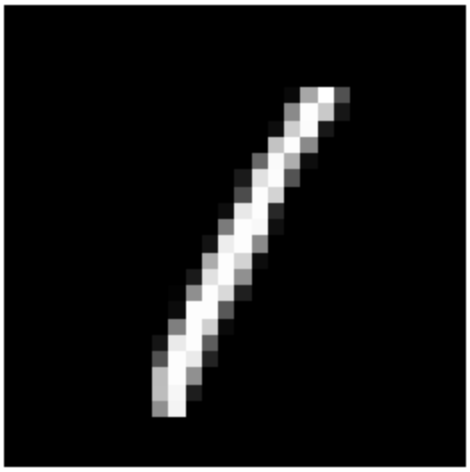
Pred: 8 | True: 8 | Confidence: 0.9996



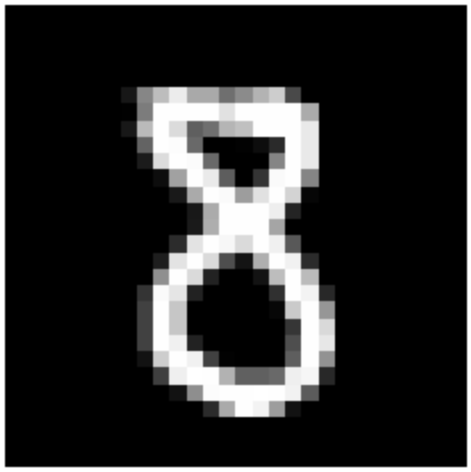
Pred: 3 | True: 3 | Confidence: 1.0000

Pred: 9 | True: 9 | Confidence: 1.0000

Pred: 1 | True: 1 | Confidence: 1.0000

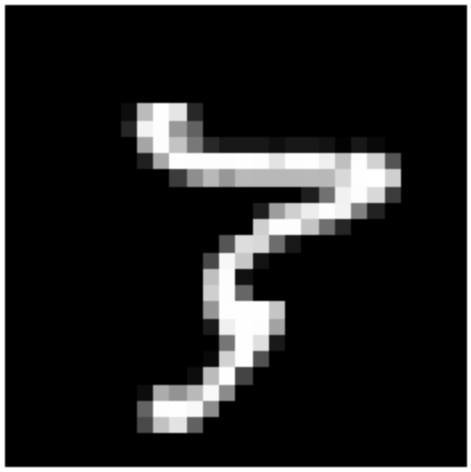


Pred: 8 | True: 8 | Confidence: 0.9966



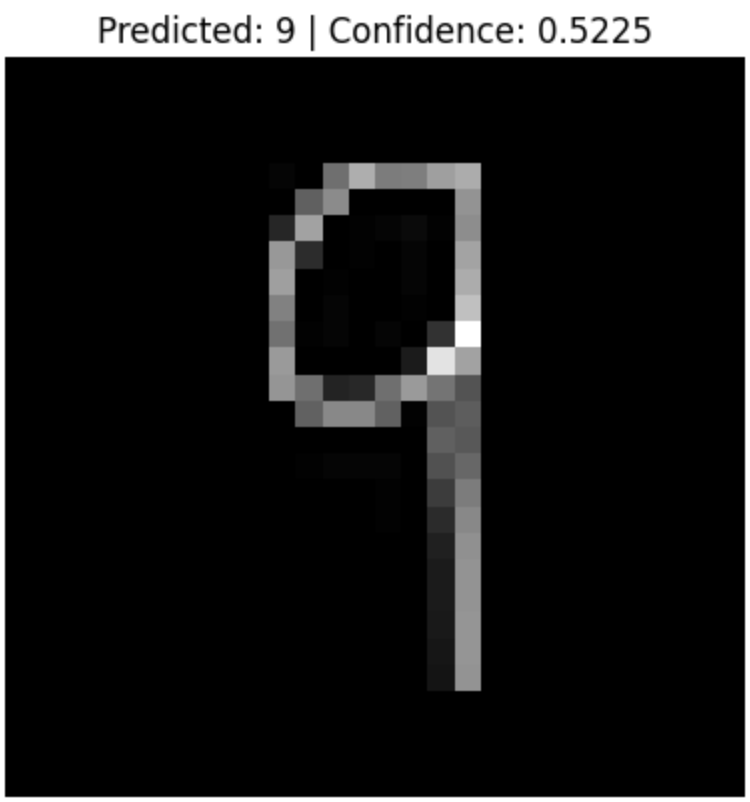
Pred: 2 | True: 2 | Confidence: 1.0000

Pred: 3 | True: 3 | Confidence: 0.8457



In [12]:
```
## Bonus

from PIL import Image, ImageOps

# Load image
img = Image.open('bonus.png').convert('L')

# Convert to numpy for quick analysis
img_array = np.array(img)

## MNIST expects dark background and bright digit so we will invert the colo
if np.mean(img_array) > 127:  # If the average pixel value is high, we assum
    img = ImageOps.invert(img)

## Convert to array again after potential inversion
img_array = np.array(img)

## Threshold to isolate the digit
## Values above threshold are treated as part of the digit
threshold = 50
coords = np.argwhere(img_array > threshold)

if len(coords) == 0:
    raise ValueError("No digit detected in the image. Try a clearer image.")

## Crop tightly around the digit
y0, x0 = coords.min(axis=0)
y1, x1 = coords.max(axis=0) + 1
digit = img_array[y0:y1, x0:x1]

## Resize while keeping aspect ratio
h, w = digit.shape
scale = 20.0 / max(h, w)   # fit digit into a 20x20 box like MNIST
new_h = max(1, int(round(h * scale)))
new_w = max(1, int(round(w * scale)))

digit_img = Image.fromarray(digit)
digit_resized = digit_img.resize((new_w, new_h), Image.Resampling.LANCZOS)
digit_resized = np.array(digit_resized)
```

```
## Create a 28x28 black canvas and center the digit
canvas = np.zeros((28, 28), dtype=np.uint8)
y_offset = (28 - new_h) // 2
x_offset = (28 - new_w) // 2
canvas[y_offset:y_offset + new_h, x_offset:x_offset + new_w] = digit_resizec

## Normalize for the model
final_img = canvas.astype("float32") / 255.0
img_input = final_img.reshape(1, 28, 28)

## Predict
prediction = prob_model.predict(img_input, verbose=0)
predicted_label = np.argmax(prediction)
confidence = np.max(prediction)

## Show processed image
plt.imshow(final_img, cmap="gray")
plt.title(f"Predicted: {predicted_label} | Confidence: {confidence:.4f}")
plt.axis("off")
plt.show()

print("Predicted label:", predicted_label)
print("Confidence score:", confidence)
```




```
Predicted label: 9
Confidence score: 0.52253914
```